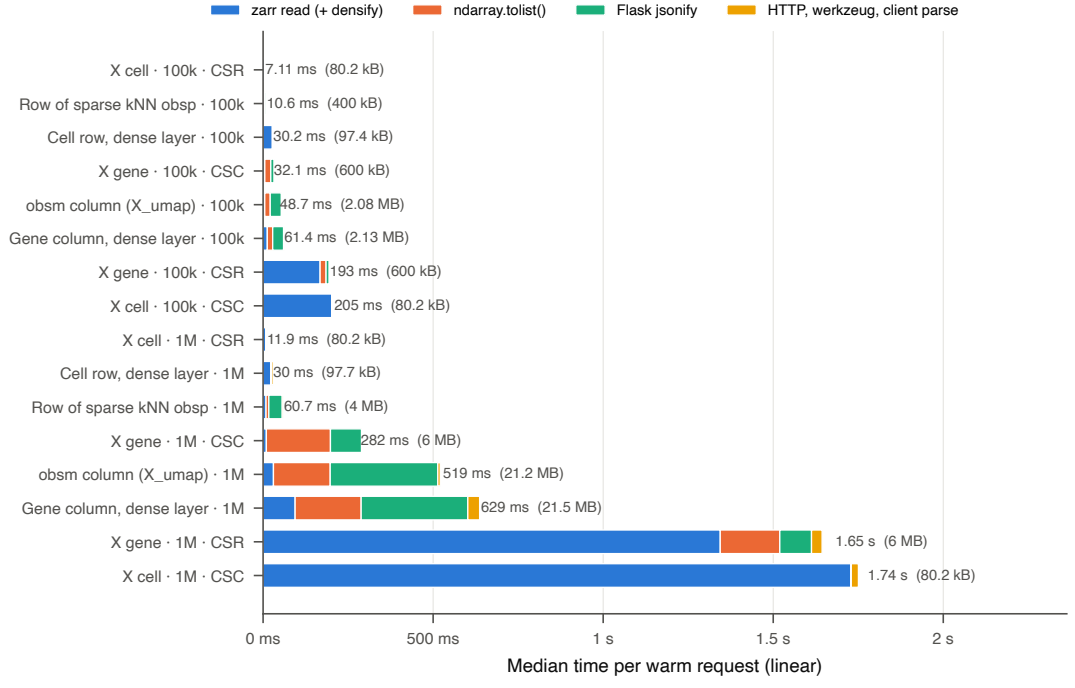
zarr read (+ densify)
ndarray.tolist()
Flask jsonify
HTTP, werkzeug, client parse
X cell · 100k · CSR
Row of sparse kNN obsp · 100k
Cell row, dense layer · 100k
X gene · 100k · CSC
obsm column (X_umap) · 100k
Gene column, dense layer · 100k
X gene · 100k · CSR
X cell · 100k · CSC
X cell · 1M · CSR
Cell row, dense layer · 1M
Row of sparse kNN obsp · 1M
X gene · 1M · CSC
obsm column (X_umap) · 1M
Gene column, dense layer · 1M
X gene · 1M · CSR
X cell · 1M · CSC
7.11 ms (80.2 kB)
10.6 ms (400 kB)
30.2 ms (97.4 kB)
32.1 ms (600 kB)
48.7 ms (2.08 MB)
61.4 ms (2.13 MB)
193 ms (600 kB)
205 ms (80.2 kB)
11.9 ms (80.2 kB)
30 ms (97.7 kB)
60.7 ms (4 MB)
282 ms (6 MB)
519 ms (21.2 MB)
629 ms (21.5 MB)
1.65 s (6 MB)
1.74 s (80.2 kB)
0 ms
500 ms
1 s
1.5 s
2 s
Median time per warm request (linear)